Type MS · Syntax, logic and runtime errors

Answer 1 of 6 · Question 1 part b.i · 3 marks

2025 JUN · Paper 02 · extract 1 of 5 · find part b.i in this mark scheme extract

1(a)	A1	**The only correct answer is B** ***A*** *is not correct because for is used in iteration* ***C*** *is not correct because new is used to declare an object* ***D*** *is not correct because while is used in repetition*		**1**

Question	mp	Answer	Additional Guidance	Mark
1(b)(i)	B1 B2 B3	Award one mark for each correct cell:		**3**

Description	Syntax error	Logic error	Runtime error
The program does not produce the expected output.		✔	
The program does not translate.	✔		
The program crashes during execution.			✔

Type MS · Syntax, logic and runtime errors

Answer 1 of 6 · Question 1 part b.i · 3 marks

2025 JUN · Paper 02 · extract 2 of 5 · find part b.i in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional guidance	Mark
1(b)(ii)	B4	Line 15 - : added to end of line		
	B5	Line 20 – 0 changed to 1		
	B6	Line 22 – missing comma added between string and greenCount		3

```
# Q01bii

# initialise counters
ballCount = 0
greenCount = 0
redCount = 0

ballArray = ["green", "red", "green", "green", "red", "green","green", "green",
             "green", "green", "green", "red", "green", "red", "green", "red",
             "red", "green", "green", "red", "red", "green", "green", "red",
             "green", "red", "red", "green", "green", "red", "green", "green",
             "red", "green", "green", "green", "red", "green", "red","green"]

# iterate over ball array
for ball in ballArray:
    ballCount = ballCount + 1
    if ball == "green":
        greenCount = greenCount +1
    else:
        redCount = redCount + 1

print(" Green: ", greenCount, "balls \n", "Red: ", redCount,
      "balls \n", "Total: ", ballCount, "balls")
```

2 of 23

Type MS · Syntax, logic and runtime errors

Answer 1 of 6 · Question 1 part b.i · 3 marks

2025 JUN · Paper 02 · extract 3 of 5 · find part b.i in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional guidance	Mark
1(c)(i)	C1	Award one mark for: • To store a value that must not be changed/prevent accidental change of value • To simplify code maintenance/so that if the value must be updated the change only has to made in a single place • To avoid inconsistency error of specifying same value multiple times • To make code more readable/give meaningful names to values	Accidental change cannot refer to 'user'	1

Question	mp	Answer	Additional guidance	Mark
1(c)(ii)	C2 C3	Award **two** marks for a linked description such as: • A variable stores a single value (1) whereas an array stores multiple values (1)		2

Question	mp	Answer	Additional guidance	Mark
1(d)(i)	D1	Award **one** mark for: • getWages (1)	Accept 'main' Accept: def getWages def getWages (inSales, inAge) line 7 (Python)	1

3 of 23

Type MS · Syntax, logic and runtime errors

Answer 1 of 6 · Question 1 part b.i · 3 marks

2025 JUN · Paper 02 · extract 4 of 5 · find part b.i in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional guidance	Mark
1(d)(ii)	D2	Award **one** mark for: • and (1)	Do not accept line numbers Accept: &&	1

Question	mp	Answer	Additional guidance	Mark
1(d)(iii)	D3	Award **one** mark for any of the following: • sales (1) • wages (1) • inSales (1)	Do not accept line numbers	1

Question	mp	Answer	Additional guidance	Mark
1(d)(iv)	D4	Award **one** mark for any of the following: • inSales (1) • inAge (1)	Do not accept line numbers	1

4 of 23

Type MS · Syntax, logic and runtime errors

Answer 1 of 6 · Question 1 part b.i · 3 marks

2025 JUN · Paper 02 · extract 5 of 5 · find part b.i in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional Guidance	Mark

Type MS · Syntax, logic and runtime errors

Answer 2 of 6 · Question 1 part b · 1 mark

2019 JUN · Paper 02 · extract 1 of 10 · find part b in this mark scheme extract

1(a)	**The only correct answer is C** A is not correct because Boolean only has values True and False B is not correct because it only allows for one character D is not correct because it is used for mixed text and numeric data		(1)

Type MS · Syntax, logic and runtime errors

Answer 2 of 6 · Question 1 part b · 1 mark

2019 JUN · Paper 02 · extract 2 of 10 · find part b in this mark scheme extract

1(b)	(an error that) produces incorrect /unintended output (1)	Accept an example of an explained logic error	(1)

Type MS · Syntax, logic and runtime errors

Answer 2 of 6 · Question 1 part b · 1 mark

2019 JUN · Paper 02 · extract 3 of 10 · find part b in this mark scheme extract

1(d)	Any one from: 1. explain how the program works (1) 2. improve readability of the program (1) 3. make code more understandable (1)		(1)

Type MS · Syntax, logic and runtime errors

Answer 2 of 6 · Question 1 part b · 1 mark

2019 JUN · Paper 02 · extract 4 of 10 · find part b in this mark scheme extract

1(e)	1. Global variables created in the main program / can be accessed from any part of the program / are in scope throughout the program (1) 2. Local variables are declared/created inside a function/blocked code (indented, brackets) and are only accessible there / are only in scope within the function in which they're declared (1)		(2)

Type MS · Syntax, logic and runtime errors

Answer 2 of 6 · Question 1 part b · 1 mark

2019 JUN · Paper 02 · extract 5 of 10 · find part b in this mark scheme extract

Question	Answer	Additional Guidance	Mark

Type MS · Syntax, logic and runtime errors

Answer 2 of 6 · Question 1 part b · 1 mark

2019 JUN · Paper 02 · extract 6 of 10 · find part b in this mark scheme extract

<table>
<tr>
<td>1(g)</td>
<td>For up to two tests
Award 1 mark for an appropriate validation test and 1 mark for an item of test data that would fail only the given test

<table>
<tr><th>Validation test</th><th>Erroneous data</th></tr>
<tr><td>Are the first two characters letters/upper case letters?</td><td>1234CCC</td></tr>
<tr><td>Are the third- and fourth-characters numerical digits?</td><td>CCCCBBB</td></tr>
<tr><td>Are the last three characters letters/upper case letters?</td><td>CC12345</td></tr>
</table>
</td>
<td>Each test should target a different feature of one of the requirements

Exactly 7 characters (in the stem)
First two characters are letters / upper case letters
Characters 3 & 4 are numeric
Last three characters are letters / upper case letters

</td>
<td>(4)</td>
</tr>
</table>

Question	Answer	Additional Guidance	Mark

Type MS · Syntax, logic and runtime errors

Answer 2 of 6 · Question 1 part b · 1 mark

2019 JUN · Paper 02 · extract 7 of 10 · find part b in this mark scheme extract

1(c)	**C#** 1. Semicolon at the end of line 9 [maxValue=5;] (1) 2. Consistent capitalisation of 'count' on lines 10, 11, 13 and 14 (1) 3. constantValue needs to be defined and given the value 7 / replaced by value 7 (1) **Java** 1. Semicolon at the end of line 8 [maxValue = 5;] (1) 2. Consistent capitalisation of 'count' on lines 7, 9, 11 and 12 (1) 3. constantValue needs to be defined and given the value 7 / replaced by 7 (1) **Python** 1. Colon needed at end of line 7 [count, maxValue):] (1) 2. Consistent capitalisation of 'count' on line 3, 7, 8 and 9 (1) 3. constantValue needs to be defined and given the value 7 / replaced by 7 (1)	Candidates are required to open the file Q01c in the code editor. Amended code should be saved as Q01cFINISHED. Line numbers may have been changed by corrections to code (e.g. insertion of constantValue definition)	(3)

Question	Answer	Additional Guidance	Mark

Type MS · Syntax, logic and runtime errors

Answer 2 of 6 · Question 1 part b · 1 mark

2019 JUN · Paper 02 · extract 8 of 10 · find part b in this mark scheme extract

<table>
<tr>
<td>1(f)</td>
<td>Award 1 mark for each correct pair of relational operators and 1 mark for correct Boolean operator for each condition.

<table>
<tr><th>Condition</th><th>Coding</th></tr>
<tr><td rowspan="2">Shop income more than £5000 or assistant sold at least 10 pairs</td><td>shopIncome > 5000; assistantSales >= 10 (1)</td></tr>
<tr><td>correct OR operator (1)</td></tr>
<tr><td rowspan="2">Shop income at least £2000 and assistant sold at least 5 pairs</td><td>shopIncome >= 2000; assistantSales >= 5 (1)</td></tr>
<tr><td>correct AND operator (1)</td></tr>
</table>

Accept >9, >1999 and >4 for respective >= comparisons</td>
<td>Candidates are required to open the file Q01f in the code editor.
Amended code should be saved as Q01fFINISHED.
Do not penalise candidates who attempt more than the stated requirements.
Do not award Boolean operator mark for single | or & as these are not correct.</td>
<td>(4)</td>
</tr>
</table>

Type MS · Syntax, logic and runtime errors

Answer 2 of 6 · Question 1 part b · 1 mark

2019 JUN · Paper 02 · extract 9 of 10 · find part b in this mark scheme extract

Question	Answer	Additional Guidance	Mark

Type MS · Syntax, logic and runtime errors

Answer 2 of 6 · Question 1 part b · 1 mark

2019 JUN · Paper 02 · extract 10 of 10 · find part b in this mark scheme extract

Question	Answer	Additional Guidance	Mark

Type MS · Syntax, logic and runtime errors

Answer 3 of 6 · Question 1 part b · 3 marks

2021 NOV · Paper 02 · extract 1 of 9 · find part b in this mark scheme extract

1 (a)	A1	**Award 1 mark for any of:** • Comments (1) • Indentation (1) • Meaningful variable/constant/subprogram names (1) • White space (1)		(1)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Syntax, logic and runtime errors

Answer 3 of 6 · Question 1 part b · 3 marks

2021 NOV · Paper 02 · extract 2 of 9 · find part b in this mark scheme extract

1 (b)	B1 B2 B3	**Award** 1 mark for each column: **Error description** / **Logic** / **Syntax** / **Runtime** Divide by 0 / / / ✓ Use x instead of * multiply / / ✓ / Subtract 10 from 2 instead of 2 from 10 / ✓ / /		(3)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Syntax, logic and runtime errors

Answer 3 of 6 · Question 1 part b · 3 marks

2021 NOV · Paper 02 · extract 3 of 9 · find part b in this mark scheme extract

1 (c)	C1	**The only correct answer is D** *A is not correct because as it is a data structure* *B is not correct because as there is no arrow looping back to the condition* *C is not correct because as the flowchart does not include an operator*		(1)

Type MS · Syntax, logic and runtime errors

Answer 3 of 6 · Question 1 part b · 3 marks

2021 NOV · Paper 02 · extract 4 of 9 · find part b in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Syntax, logic and runtime errors

Answer 3 of 6 · Question 1 part b · 3 marks

2021 NOV · Paper 02 · extract 5 of 9 · find part b in this mark scheme extract

1 (d)	**Award 1 mark for each of:** D1 = changed to == (1) D2 Condition changed to less than / final two print statements swapped (1) D3 else changed to else (1)		(3)

Code examples

C#

```
if (number1 == number2)
{
    Console.WriteLine("The numbers are equal");
}
else if (number1 < number2)
{
    Console.WriteLine("The highest number is " + number2 + " and the lowest number is " + number1);
}
else
{
```

Java

```
if (number1 == number2)
{
    System.out.print("Numbers are equal");
}
else if (number1 < number2)
{
    System.out.print("The highest number is " + number2 + " and the lowest number is " +number1);
}
else
```

Python

```
if number1 == number2:
    print("Numbers are equal")
elif number1 < number2:
    print("The highest number is", number2, "and the lowest number is", number1)
else:
```

Type MS · Syntax, logic and runtime errors

Answer 3 of 6 · Question 1 part b · 3 marks

2021 NOV · Paper 02 · extract 6 of 9 · find part b in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Syntax, logic and runtime errors

Answer 3 of 6 · Question 1 part b · 3 marks

2021 NOV · Paper 02 · extract 7 of 9 · find part b in this mark scheme extract

1 (e)	**Award 3 marks for:**			
	E1	Prompt for a number to be input (1)		
	E2	Input stored in an integer variable (1)		
	E3	Loop used (1)		
	E4	Loop iterates only for the values 1 – 12 (1)		
	E5	Display is correct when program executes (1)		(5)

Code examples

C#

```
// Print prompt and get number from user
Console.Write("Enter the number: ");
int num = Convert.ToInt32(Console.ReadLine());

// Create loop to display the table
for (int count = 1; count <= 12; count++)
{
    Console.WriteLine(count + " x " + num + " is " + count * num);
}
Console.ReadLine();
```

Java

```
// Print prompt and get number from user
Scanner input = new Scanner(System.in);
System.out.print("Enter the number ");
int num = input.nextInt();

// Create loop to display the table
for (int count = 1; count <=12; count++)
{
    System.out.println(count + " x " + num + " is " + count * num);
}
```

Type MS · Syntax, logic and runtime errors

Answer 3 of 6 · Question 1 part b · 3 marks

2021 NOV · Paper 02 · extract 8 of 9 · find part b in this mark scheme extract

Python	

```python
# Print prompt and get number from user
num = int(input("Enter the number: "))

# Create loop to display the table
for count in range(1, 13):
    print(count, "x" , num , "is" , count * num)
```

Type MS · Syntax, logic and runtime errors

Answer 3 of 6 · Question 1 part b · 3 marks

2021 NOV · Paper 02 · extract 9 of 9 · find part b in this mark scheme extract

Question	mp		Additional Guidance	Mark

Type MS · Syntax, logic and runtime errors

Answer 4 of 6 · Question 1 part e · 1 mark

2022 JUN · Paper 02 · extract 1 of 9 · find part e in this mark scheme extract

1 (a)	A1	**The only correct answer is D** A is not correct because the value can be used more than once B is not correct because the value does not always have to be input C is not correct because the value does not always have to be used in a calculation		(1)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Syntax, logic and runtime errors

Answer 4 of 6 · Question 1 part e · 1 mark

2022 JUN · Paper 02 · extract 2 of 9 · find part e in this mark scheme extract

1 (b)	B1	**The only correct answer is B** A is not correct because this would not aid readability of the code C is not correct because this would not aid readability of the code D is not correct because this would not aid readability of the code		(1)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Syntax, logic and runtime errors

Answer 4 of 6 · Question 1 part e · 1 mark

2022 JUN · Paper 02 · extract 3 of 9 · find part e in this mark scheme extract

<table>
<tr><td>1 (c)</td><td>C1
C2</td><td>Award 1 mark for each of:
<table><tr><th>Data type</th><th>Example value</th></tr><tr><td>char</td><td>An example of any single letter, number or symbol</td></tr><tr><td>real</td><td>An example of a number with a decimal point</td></tr></table></td><td>Accept quotes around the character

Real number does not need a number after the decimal point to be awarded the mark</td><td>(2)</td></tr>
</table>

Question	mp	Answer	Additional Guidance	Mark

Type MS · Syntax, logic and runtime errors

Answer 4 of 6 · Question 1 part e · 1 mark

2022 JUN · Paper 02 · extract 4 of 9 · find part e in this mark scheme extract

1 (d)	D1	**Award up to 2 marks for a description such as:** • Boundary testing uses the most extreme valid data / minimum and maximum valid values (1) whereas erroneous testing uses invalid data (1)		(2)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Syntax, logic and runtime errors

Answer 4 of 6 · Question 1 part e · 1 mark

2022 JUN · Paper 02 · extract 5 of 9 · find part e in this mark scheme extract

1 (e)	E1	Logic/logical		(1)

Type MS · Syntax, logic and runtime errors

Answer 4 of 6 · Question 1 part e · 1 mark

2022 JUN · Paper 02 · extract 6 of 9 · find part e in this mark scheme extract

Question	mp		Additional Guidance	Mark

Type MS · Syntax, logic and runtime errors

Answer 4 of 6 · Question 1 part e · 1 mark

2022 JUN · Paper 02 · extract 7 of 9 · find part e in this mark scheme extract

1 (f)	**Award 1 mark for each of:**			
	F1	Bracket added and no other amendments made to the line (1)		
	F2	== becomes = (1)		
	F3	" moved to after **is** and before , OR , area changed to +str(area) (1) Must still include the print statement		(3)
Code examples				
Python	3 length = float(input("Enter the first number ") 4 5 area = length * length 6 7 print("The area of the square is" , area)			

Question	mp	Answer	Additional Guidance	Mark

Type MS · Syntax, logic and runtime errors

Answer 4 of 6 · Question 1 part e · 1 mark

2022 JUN · Paper 02 · extract 8 of 9 · find part e in this mark scheme extract

1 (g)	**Award 1 mark for each of:**			
	G1	At least 1 condition correct (1)		
	G2	At least 1 > or < condition and message match (1)		
	G3	At least 1 output includes letter and stored letter with an appropriate message (1)		
	G4	All conditions and outputs correct (1)		(4)
Code examples				
Python				

```
# Amend the code by completing the if statement
if letter > storedLetter:
    print(letter + " is later in the alphabet than " + storedLetter)
elif letter == storedLetter:
    print(letter + " is the same as " + storedLetter)
else:
    print(letter + " is earlier in the alphabet than " + storedLetter)
```

Type MS · Syntax, logic and runtime errors

Answer 4 of 6 · Question 1 part e · 1 mark

2022 JUN · Paper 02 · extract 9 of 9 · find part e in this mark scheme extract

Question	mp		Additional Guidance	Mark

Type MS · Syntax, logic and runtime errors

Answer 5 of 6 · Question 2 part b.i · 1 mark

2024 JUN · Paper 02 · extract 1 of 14 · find part b.i in this mark scheme extract

2(a)(i)	A1	**The only correct answer is A** ***B*** *is not correct because* ***computational thinking*** *are skills used in creating algorithms and programs* ***C*** *is not correct because* ***decomposition*** *is breaking down a problem into smaller parts* ***D*** *is not correct because* ***pseudocode*** *is a notation for expressing an algorithm*		**1**

Type MS · Syntax, logic and runtime errors

Answer 5 of 6 · Question 2 part b.i · 1 mark

2024 JUN · Paper 02 · extract 2 of 14 · find part b.i in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Syntax, logic and runtime errors

Answer 5 of 6 · Question 2 part b.i · 1 mark

2024 JUN · Paper 02 · extract 3 of 14 · find part b.i in this mark scheme extract

2(a)(ii)	A2	Award **one** mark for: • Variable (1)	Allow parameter Ignore local/global	**1**

Question	mp	Answer	Additional Guidance	Mark

Type MS · Syntax, logic and runtime errors

Answer 5 of 6 · Question 2 part b.i · 1 mark

2024 JUN · Paper 02 · extract 4 of 14 · find part b.i in this mark scheme extract

2(b)(i)	B1	Award **one** mark for any of the following: • An instruction is impossible to execute (1) • The program stops executing **unexpectedly** / **crashes** (1)	Do not award an example, as that is the next question Don't award repeat of question e.g. "Stopping during execution" without clarifying reason or that stop was unexpected	**1**

Question	mp	Answer	Additional Guidance	Mark

Type MS · Syntax, logic and runtime errors

Answer 5 of 6 · Question 2 part b.i · 1 mark

2024 JUN · Paper 02 · extract 5 of 14 · find part b.i in this mark scheme extract

2(b)(ii)	B2	Award **one** mark for any of the following: • Division by zero (1) • Accessing memory outside the range of an array (1) • Attempting an operation on incompatible types (1) • File not found (1) • An exception is raised that cannot be/is not handled (1)	Award other examples, if they can be mapped to a bullet	**1**

Type MS · Syntax, logic and runtime errors

Answer 5 of 6 · Question 2 part b.i · 1 mark

2024 JUN · Paper 02 · extract 6 of 14 · find part b.i in this mark scheme extract

Type MS · Syntax, logic and runtime errors

Answer 5 of 6 · Question 2 part b.i · 1 mark

2024 JUN · Paper 02 · extract 7 of 14 · find part b.i in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

Type MS · Syntax, logic and runtime errors

Answer 5 of 6 · Question 2 part b.i · 1 mark

2024 JUN · Paper 02 · extract 8 of 14 · find part b.i in this mark scheme extract

2(c)		Award **one** mark for each of:		
	C1	Add or remove a semicolon on initialisation of the length variable (1)	• `int length = 4;` • `int length = 4;` • `length = 4`	
	C2	Add a comma between the arguments to the subprogram call to generate a random number (1)	• `width = randGenerator.Next (1, 6);` • `width = randGenerator.nextInt(1, 6);` • `width = random.randint (1, 5)`	
	C3	Replace modulus operator (%) with multiplication operator (*) (1)	• `perimeter = (2 * width) + (2 * length);` • `perimeter = (2 * width) + (2 * length);` • `perimeter = (2 * width) + (2 * length)`	**3**

Type MS · Syntax, logic and runtime errors

Answer 5 of 6 · Question 2 part b.i · 1 mark

2024 JUN · Paper 02 · extract 9 of 14 · find part b.i in this mark scheme extract

C#

```
// Q02cFINISHED

int width = 0;
int length = 4;
int perimeter = 0;
Random randGenerator = new Random();

width = randGenerator.Next (1, 6);
perimeter = (2 * width) + (2 * length);

Console.WriteLine (width.ToString() + " " +
                   length.ToString() + " " +
                   perimeter.ToString());
```

Type MS · Syntax, logic and runtime errors

Answer 5 of 6 · Question 2 part b.i · 1 mark

2024 JUN · Paper 02 · extract 10 of 14 · find part b.i in this mark scheme extract

Java

```java
// Q02cFINISHED

import java.util.Random;

public class Q02cFINISHED
{
    public static void main (String[] args)
    {
        int width = 0;
        int length = 4;
        int perimeter = 0;
        Random randGenerator = new Random();

        width = randGenerator.nextInt(1, 6);
        perimeter = (2 * width) + (2 * length);

        System.out.println (String.valueOf(width) + " " +
                            String.valueOf(length) + " " +
                            String.valueOf(perimeter));
    }
}
```

Type MS · Syntax, logic and runtime errors

Answer 5 of 6 · Question 2 part b.i · 1 mark

2024 JUN · Paper 02 · extract 11 of 14 · find part b.i in this mark scheme extract

Python

```python
# Q02cFINISHED

import random

width = 0
length = 4
perimeter = 0

width = random.randint (1, 5)
perimeter = (2 * width) + (2 * length)

print (width, length, perimeter)

```

Type MS · Syntax, logic and runtime errors

Answer 5 of 6 · Question 2 part b.i · 1 mark

2024 JUN · Paper 02 · extract 12 of 14 · find part b.i in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Syntax, logic and runtime errors

Answer 5 of 6 · Question 2 part b.i · 1 mark

2024 JUN · Paper 02 · extract 13 of 14 · find part b.i in this mark scheme extract

<table>
<tr>
<td>2(d)</td>
<td>D1
D2
D3
D4</td>
<td>
<table>
<tr><td>Erroneous</td><td>Award one mark for any of the following up to a maximum of two marks:
Contains incorrect data type (1)
<2 letters (1)
9+ letters (1)</td></tr>
<tr><td>Normal</td><td>A string between 2 and 8 in length that includes only letters (1)</td></tr>
<tr><td>Boundary</td><td>Award one mark for any of the following:
A string of exactly 8 letters (1)
A string of exactly 2 letters (1)</td></tr>
</table>
</td>
<td>Award any correct example (not description)
Do not award an empty cell as equivalent to empty string
Do not accept same test twice for erroneous</td>
<td>4</td>
</tr>
</table>

Type MS · Syntax, logic and runtime errors

Answer 5 of 6 · Question 2 part b.i · 1 mark

2024 JUN · Paper 02 · extract 14 of 14 · find part b.i in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Syntax, logic and runtime errors

Answer 6 of 6 · Question 2 part b.ii · 1 mark

2024 JUN · Paper 02 · extract 1 of 14 · find part b.ii in this mark scheme extract

2(a)(i)	A1	**The only correct answer is A** ***B*** *is not correct because* ***computational thinking*** *are skills used in creating algorithms and programs* ***C*** *is not correct because* ***decomposition*** *is breaking down a problem into smaller parts* ***D*** *is not correct because* ***pseudocode*** *is a notation for expressing an algorithm*		**1**

Type MS · Syntax, logic and runtime errors

Answer 6 of 6 · Question 2 part b.ii · 1 mark

2024 JUN · Paper 02 · extract 2 of 14 · find part b.ii in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Syntax, logic and runtime errors

Answer 6 of 6 · Question 2 part b.ii · 1 mark

2024 JUN · Paper 02 · extract 3 of 14 · find part b.ii in this mark scheme extract

2(a)(ii)	A2	Award **one** mark for: • Variable (1)	Allow parameter Ignore local/global	**1**

Question	mp	Answer	Additional Guidance	Mark

Type MS · Syntax, logic and runtime errors

Answer 6 of 6 · Question 2 part b.ii · 1 mark

2024 JUN · Paper 02 · extract 4 of 14 · find part b.ii in this mark scheme extract

2(b)(i)	B1	Award **one** mark for any of the following: • An instruction is impossible to execute (1) • The program stops executing **unexpectedly** / **crashes** (1)	Do not award an example, as that is the next question Don't award repeat of question e.g. "Stopping during execution" without clarifying reason or that stop was unexpected	**1**

Question	mp	Answer	Additional Guidance	Mark

Type MS · Syntax, logic and runtime errors

Answer 6 of 6 · Question 2 part b.ii · 1 mark

2024 JUN · Paper 02 · extract 5 of 14 · find part b.ii in this mark scheme extract

2(b)(ii)	B2	Award **one** mark for any of the following: • Division by zero (1) • Accessing memory outside the range of an array (1) • Attempting an operation on incompatible types (1) • File not found (1) • An exception is raised that cannot be/is not handled (1)	Award other examples, if they can be mapped to a bullet	**1**

Type MS · Syntax, logic and runtime errors

Answer 6 of 6 · Question 2 part b.ii · 1 mark

2024 JUN · Paper 02 · extract 6 of 14 · find part b.ii in this mark scheme extract

Type MS · Syntax, logic and runtime errors

Answer 6 of 6 · Question 2 part b.ii · 1 mark

2024 JUN · Paper 02 · extract 7 of 14 · find part b.ii in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

Type MS · Syntax, logic and runtime errors

Answer 6 of 6 · Question 2 part b.ii · 1 mark

2024 JUN · Paper 02 · extract 8 of 14 · find part b.ii in this mark scheme extract

2(c)		Award **one** mark for each of:		
	C1	Add or remove a semicolon on initialisation of the length variable (1)	• `int length = 4;` • `int length = 4;` • `length = 4`	
	C2	Add a comma between the arguments to the subprogram call to generate a random number (1)	• `width = randGenerator.Next (1, 6);` • `width = randGenerator.nextInt(1, 6);` • `width = random.randint (1, 5)`	
	C3	Replace modulus operator (%) with multiplication operator (*) (1)	• `perimeter = (2 * width) + (2 * length);` • `perimeter = (2 * width) + (2 * length);` • `perimeter = (2 * width) + (2 * length)`	**3**

Type MS · Syntax, logic and runtime errors

Answer 6 of 6 · Question 2 part b.ii · 1 mark

2024 JUN · Paper 02 · extract 9 of 14 · find part b.ii in this mark scheme extract

C#

```
// Q02cFINISHED

int width = 0;
int length = 4;
int perimeter = 0;
Random randGenerator = new Random();

width = randGenerator.Next (1, 6);
perimeter = (2 * width) + (2 * length);

Console.WriteLine (width.ToString() + " " +
                   length.ToString() + " " +
                   perimeter.ToString());
```

Type MS · Syntax, logic and runtime errors

Answer 6 of 6 · Question 2 part b.ii · 1 mark

2024 JUN · Paper 02 · extract 10 of 14 · find part b.ii in this mark scheme extract

Java

```java
// Q02cFINISHED

import java.util.Random;

public class Q02cFINISHED
{
    public static void main (String[] args)
    {
        int width = 0;
        int length = 4;
        int perimeter = 0;
        Random randGenerator = new Random();

        width = randGenerator.nextInt(1, 6);
        perimeter = (2 * width) + (2 * length);

        System.out.println (String.valueOf(width) + " " +
                            String.valueOf(length) + " " +
                            String.valueOf(perimeter));
    }
}
```

Type MS · Syntax, logic and runtime errors

Answer 6 of 6 · Question 2 part b.ii · 1 mark

2024 JUN · Paper 02 · extract 11 of 14 · find part b.ii in this mark scheme extract

Python

```python
# Q02cFINISHED

import random

width = 0
length = 4
perimeter = 0

width = random.randint (1, 5)
perimeter = (2 * width) + (2 * length)

print (width, length, perimeter)

```

Type MS · Syntax, logic and runtime errors

Answer 6 of 6 · Question 2 part b.ii · 1 mark

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Question	mp	Answer	Additional Guidance	Mark

Type MS · Syntax, logic and runtime errors

Answer 6 of 6 · Question 2 part b.ii · 1 mark

2024 JUN · Paper 02 · extract 13 of 14 · find part b.ii in this mark scheme extract

<table>
<tr>
<td>2(d)</td>
<td>D1
D2
D3
D4</td>
<td>
<table>
<tr><td>Erroneous</td><td>Award one mark for any of the following up to a maximum of two marks:
Contains incorrect data type (1)
<2 letters (1)
9+ letters (1)</td></tr>
<tr><td>Normal</td><td>A string between 2 and 8 in length that includes only letters (1)</td></tr>
<tr><td>Boundary</td><td>Award one mark for any of the following:
A string of exactly 8 letters (1)
A string of exactly 2 letters (1)</td></tr>
</table>
</td>
<td>Award any correct example (not description)
Do not award an empty cell as equivalent to empty string
Do not accept same test twice for erroneous</td>
<td>4</td>
</tr>
</table>

Type MS · Syntax, logic and runtime errors

Answer 6 of 6 · Question 2 part b.ii · 1 mark

2024 JUN · Paper 02 · extract 14 of 14 · find part b.ii in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark